Fine-Tuning a Language Model for Bird Conservation Report Drafting

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Abstract

Preparing a conservation report from incomplete biodiversity information requires more than a fluent summary: the report must preserve the distinction between available evidence and unanswered questions. We investigated whether parameter-efficient fine-tuning could help a language model perform this task within EcoPath, a geospatial bird-intelligence platform. We adapted Qwen2.5-7B-Instruct using QLoRA on 85 synthetic examples and compared it with the unadapted model on 15 held-out bird species. Inputs contained GBIF-linked species metadata but lacked verified conservation status, population trend, population estimate, habitat, and range fields. The adapted model produced valid, schema-compliant JSON for all 15 species, whereas the unadapted model returned Markdown prose and passed neither structural check. Mean output length decreased from 590 to 335 words. Supplementary automated evaluation also favoured the adapted responses, with mean grounding scores of 5.00 versus 2.33 and reviewer-usefulness scores of 4.73 versus 1.87 on a five-point scale. Illustrative cases showed a shift from unsupported threat descriptions to explicit acknowledgement of missing information. The findings demonstrate adaptation to a structured drafting task under sparse inputs. Whether this behaviour represents reliable evidence use beyond the learned reporting pattern remains unresolved, particularly in comparison with stronger prompting and constrained generation. EcoPath provides a setting for investigating that distinction while retaining human responsibility for conservation interpretation.

Keywords: ecological AI; biodiversity data platform; QLoRA; bird conservation; report drafting; structured generation; missing evidence.

1 Introduction

Conservation reporting depends on assembling information that differs in its source, coverage, and relevance to the question being asked. A species may have an identifiable taxonomic record and a traceable occurrence source while still lacking the population, habitat, or threat evidence needed for an assessment. For bird-related information systems, the practical challenge is therefore not simply to retrieve records, but to communicate what those records support and where the evidence remains incomplete.

Citizen-science initiatives and open repositories have expanded opportunities to work with biodiversity information. The eBird framework illustrates how distributed observations can support avian research [1], while research on GBIF demonstrates that spatial bias can affect the interpretation of aggregated records [2]. These resources are valuable inputs to conservation workflows, but their availability should not be confused with sufficient evidence for every field in a report. A reporting system must preserve the distinction between observed information, unavailable information, and conclusions requiring further investigation.

Large language models (LLMs) offer a way to turn heterogeneous information into readable drafts. Castro et al. [9] demonstrated their potential for extracting and structuring ecological information from text, although performance differed across tasks and models. The same fluency that makes these systems useful can also conceal unsupported content [7]. In a conservation report, a plausible description of threats may

appear informative even when those threats were absent from the input. Useful drafting therefore requires both a consistent report structure and a response to uncertainty that readers can inspect.

There are several ways to address these requirements. Retrieval-augmented generation supplies external information to the generator [3], while output specifications define how that information should be presented. Parameter-efficient adaptation offers a further possibility: learning a reporting pattern from examples without updating the full model [4,6]. However, producing the requested structure and faithfully using evidence are different achievements. Format restrictions can also affect task performance [10]. An evaluation of report drafting must consequently distinguish successful delivery of an output format from the quality of the ecological content within it.

We examined this problem within EcoPath, a platform that combines bird species exploration, geospatial visualisation, and AI-assisted interaction. Its report-drafting module is intended to organise selected information for human review. We used QLoRA to adapt Qwen2.5-7B-Instruct [11] on examples in which major ecological fields were missing and compared its responses with those of the unadapted model. This setting allowed us to examine whether a small training dataset could establish a consistent report format and encourage explicit acknowledgement of missing information.

The study addressed two questions: (i) how does QLoRA adaptation change structural compliance on held-out species under the same reporting prompts, and (ii) what differences in missing-information handling are apparent in automated ratings and illustrative outputs? We expected the adapted model to follow the report schema more consistently. We treated the second question as exploratory because the inputs did not vary the availability of the principal ecological fields. The resulting contribution is a proof of concept for structured bird conservation drafting, together with an analysis of the distinction between learning a report template and learning to use evidence selectively.

2 Related Work

The reporting problem connects research on biodiversity information with work on language-model grounding, structured generation, and efficient adaptation. The literature provides reasons to investigate each component, but also makes clear why success on one component cannot substitute for evaluation of the others.

2.1 Biodiversity Information and Ecological AI

Sullivan et al. [1] describe how eBird supports the collection and reuse of bird observations through a shared information system. Beck et al. [2] examine a complementary issue: spatial bias in aggregated occurrence records can affect downstream analyses. For report drafting, these studies motivate retaining source context and avoiding conclusions that exceed the observations. Tuia et al. [5] further emphasise the role of ecological knowledge and interdisciplinary collaboration in machine-learning applications for conservation. We therefore place EcoPath within an evidence-inspection workflow in which automated drafting supports, rather than determines, conservation interpretation.

2.2 Language Models for Ecological Information Processing

Castro et al. [9] evaluated prompt-based LLMs for identifying, extracting, geocoding, and structuring species-distribution information. Their work demonstrates that language models can assist ecological data processing and that their performance depends on the task. This motivates a separate assessment of report structure and content in EcoPath. It also establishes prompting as an important comparator when determining whether additional model training is worthwhile.

Ecological knowledge benchmarks show why evidence organisation should be separated from judgement. Dorm et al. [12] found differences in LLM performance across species presence, range mapping, threat classification, and related tasks. Uryu [13], in a preprint examining IUCN-related species information, likewise reported differing performance on taxonomic and conservation-status questions. These studies do

not evaluate the same task as EcoPath, but they motivate its restricted scope: organising the supplied material without treating the draft as a formal conservation assessment.

2.3 Grounding and Structured Generation

Lewis et al. [3] introduced retrieval-augmented generation as a means of conditioning language generation on external information. For a report generator, such access is useful only if the output remains faithful to the retrieved material. Structure adds a separate requirement. Tam et al. [10] found that format restrictions can affect performance on reasoning tasks, showing that predictable output and successful task execution need independent measurement. EcoPath accordingly treats JSON validity as an integration measure and evidence support as a distinct question.

2.4 Parameter-Efficient Adaptation and Evaluation

LoRA reduces the number of parameters updated during adaptation by learning low-rank changes to frozen pretrained weights [4]. QLoRA extends this approach to a quantised backbone, reducing the memory needed for training [6]. These properties make adaptation relevant to small application-development teams, although they do not establish a quality or cost advantage for a particular workflow. We used the instruction-tuned Qwen2.5 family [11] to examine adaptation in the specific context of sparse-input report drafting.

Evaluating the resulting text introduces a further challenge. Zheng et al. [8] showed that LLM-based judges can approximate human preferences while exhibiting position, verbosity, and self-enhancement biases. Automated scores can therefore provide an exploratory comparison, but do not independently verify ecological statements. The present study combines deterministic format checks with supplementary judging and qualitative examples, while reserving claims about evidence fidelity for more direct evaluation.

3 Methodology

We implemented the drafting task within EcoPath and evaluated the same instruction-tuned model with and without a QLoRA adapter. The experiment comprised a synthetic training dataset, one adaptation run, and a paired comparison on 15 held-out species. This section describes the application context and the procedures underlying that comparison.

3.1 System Architecture

EcoPath separates user interaction, application orchestration, persistent data, external sources, and AI services (Figure 1; Table 1). A Next.js frontend provides species search, profiles, image-assisted identification, and map-based exploration, with MapLibre GL JS supporting visualisation. FastAPI handles requests and coordinates species-resolution and service workflows. Supabase and PostgreSQL provide persistent storage, while PostGIS supports spatial queries.

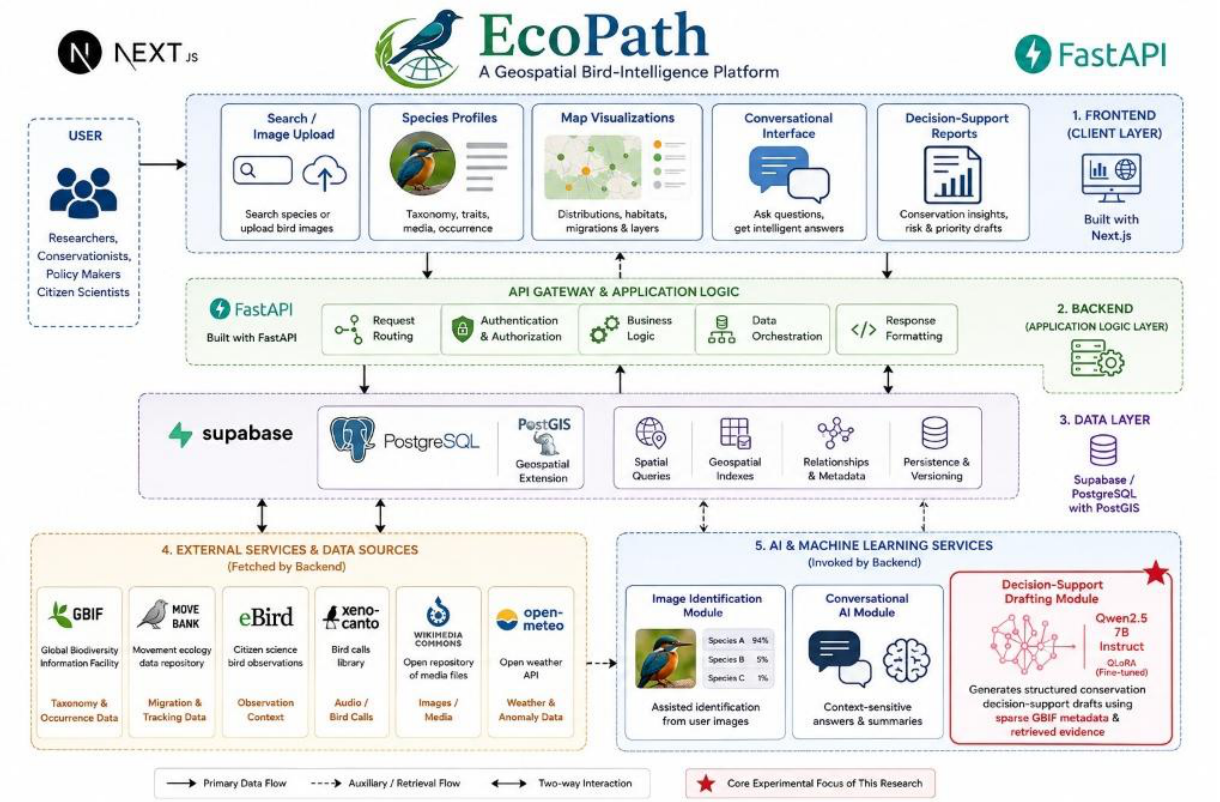


Figure 1. EcoPath architecture. The report-drafting module is the experimental focus; the remaining platform components provide the application context.

The platform connects to GBIF, Movebank, eBird, Xeno-canto, Wikimedia Commons, and Open-Meteo for different biodiversity, movement, media, and weather functions. Its AI services support identification, conversation, and report drafting. Our evaluation concerned the drafting module alone. Although several sources are accessible through the platform, the experimental inputs contained only the GBIF-linked metadata described in Section 3.3.

Table 1. EcoPath architecture and the scope of the report-generation experiment.

Layer	Responsibilities and technologies	Experimental scope
Presentation	Search, profiles, maps, report display; Next.js and MapLibre GL JS	Application context
Application services	Routing, species resolution, orchestration; FastAPI	Application context
Persistent data	Records and spatial queries; Supabase, PostgreSQL, PostGIS	Application context
External services	Biodiversity, movement, media, and weather sources	Only supplied GBIF-linked metadata evaluated
AI services	Identification, conversation, and QLoRA report drafting	Report drafting evaluated

3.2 Evidence-Bounded Decision-Support Pipeline

The workflow consisted of preparing species-linked targets, dividing the dataset into training and held-out subsets, adapting the model, and generating paired reports (Figure 2). Each model condition received the same held-out species payload and reporting prompt. We assessed the resulting responses for parseability and schema compliance, then used an automated rubric to compare their presentation and judged content quality.



Figure 2. Dataset preparation and QLoRA adaptation. The unadapted model and the resulting adapter were compared using identical inputs for 15 held-out species.

The intended evidence boundary was straightforward: organise the supplied information, retain the source reference, and acknowledge unavailable fields. The pipeline did not infer an official IUCN category or verify the draft against an independent ecological assessment. Human review was the intended use context, rather than an evaluated intervention in this experiment.

3.3 Dataset Generation and Evidence Boundary

We constructed the EcoPath Gemini-100 supervised fine-tuning dataset, comprising 100 synthetic examples for 100 bird species. Each example contained a structured species payload, retrieval queries, evidence records, and a teacher-generated report. Species identity and taxonomic metadata were resolved through GBIF. A GBIF URL provided the traceable source reference, while verified IUCN status, population trend, population estimate, habitat, and range fields were absent.

Target reports made these gaps explicit rather than supplying unsupported numerical or conservation-status claims. This design exposed the model to a consistent reporting task: express the available species context within the required structure and identify the ecological information that could not be established. Because the principal ecological fields were absent throughout, the dataset tested this sparse-input pattern without testing how behaviour changed when those fields became available.

Using a deterministic split with seed 42, we assigned 85 species examples to training and 15 to held-out evaluation. The held-out set was not used for prompt modification or checkpoint selection. There was no separate validation partition. The evaluation species are listed in Appendix A. The platform had an India-focused motivation, but the held-out list was geographically broader and was not designed to estimate performance specifically on Indian birds.

3.4 Fine-Tuning Configuration

We loaded Qwen/Qwen2.5-7B-Instruct [11] in 4-bit quantised form and trained a QLoRA adapter [6]. Updates targeted the q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, and down_proj layers. The adapter used rank 16, alpha 32, and dropout 0.05. Training proceeded for three epochs with a learning rate of 2×10^{-4} and batch size 1 with gradient accumulation, giving 33 optimisation steps (Table 2).

Table 2. Reported fine-tuning configuration.

Parameter	Value
Base model	Qwen/Qwen2.5-7B-Instruct
Quantisation	4-bit backbone
Training / held-out examples	85 / 15; split seed 42
LoRA rank / alpha / dropout	16 / 32 / 0.05
Target layers	q_proj, k_proj, v_proj, o_proj, gate_proj, up_proj, down_proj
Learning rate	2×10^{-4}
Batch size	1, with gradient accumulation
Duration	3 epochs; 33 optimisation steps
Recorded loss values	1.437 at step 10; 0.376 at step 30
Independent training runs	One reported
Validation set	None reported

We conducted one adaptation run. Training loss was recorded during optimisation, and the resulting adapter was used for the held-out comparison. No validation-based checkpoint comparison or repeated-seed analysis was conducted. The implications of this design and incomplete implementation reporting are addressed in Section 5.5.

3.5 Evaluation Protocol

We generated reports for each of the 15 held-out species in two conditions: the unadapted Qwen2.5-7B-Instruct model and the same model with the EcoPath adapter. The conditions used the same prompts, evidence payloads, and generation settings. This paired design held the input constant while examining the difference associated with the adapter under the tested configuration.

We first assessed JSON validity after removing optional Markdown fences. We then checked schema compliance for six report components: species, current conservation status, key threats, survivability assessment, mitigation actions, and sources. These checks measured whether the application could consume the response in the requested form. They did not verify the factual support of the text within each field.

We used a Gemini-family LLM judge to score all 30 responses against the supplied evidence. The five-point rubric covered format compliance, factual grounding, completeness, and reviewer usefulness. All responses received scores. These ratings were supplementary: no blinded human assessment was conducted, and independence between the judge and the synthetic-data teacher was not established.

We summarised structural outcomes as counts and proportions, report length as the mean number of words, and rubric outcomes as descriptive means. For each binary structural proportion, we calculated a two-sided 95% exact binomial interval. These intervals describe the observed proportions under a binomial assumption; they are not confidence intervals for the paired treatment difference. We did not estimate an unsupported-claim rate, field-level abstention accuracy, or citation-support rate. Qualitative analysis examined the Herring Gull and Common Murre cases presented in Appendix A.

4 Results

Adaptation changed the form and length of the generated reports and was accompanied by higher automated ratings. We first report training behaviour and structural outcomes, followed by the rubric comparison and species-level observations.

4.1 Training Behaviour

Training loss decreased from 1.437 at optimisation step 10 to 0.376 at step 30. This decline showed that the training objective decreased during the run. Held-out structural results provide the separate assessment of whether the resulting adapter followed the reporting format on species outside the training subset.

4.2 Held-Out Structural Evaluation

All 15 adapted responses were valid JSON and satisfied the schema, compared with none of the unadapted responses (Table 3). The unadapted model returned Markdown prose rather than the requested objects. Its schema failures therefore coincided with parse failures; the two structural checks described the same failure pattern rather than independent dimensions of improvement.

Table 3. Paired held-out results on 15 species. Rubric scores are automated five-point ratings.

Metric	Unadapted	QLoRA-adapted	Difference
JSON validity	0/15 (0.0%)	15/15 (100.0%)	+100.0 pp
Schema compliance	0/15 (0.0%)	15/15 (100.0%)	+100.0 pp
Mean output length	590 words	335 words	-255 words
Judge: format compliance	1.00	5.00	+4.00
Judge: factual grounding	2.33	5.00	+2.67
Judge: completeness	3.87	5.00	+1.13
Judge: reviewer usefulness	1.87	4.73	+2.86

The 95% exact binomial intervals were 0.0-21.8% for the unadapted proportion and 78.2-100.0% for the adapted proportion. Thus, although the observed outcomes were uniform within each condition, the small held-out set left substantial uncertainty about performance on additional species.

Adapted reports were shorter, averaging 335 words compared with 590 words for the unadapted responses. The judge also assigned higher mean scores to adapted reports across the four rubric dimensions (Figure 3). Grounding increased from 2.33 to 5.00, completeness from 3.87 to 5.00, and reviewer usefulness from 1.87 to 4.73. Format compliance increased from 1.00 to 5.00. These were descriptive automated ratings; variation in the individual scores was not analysed.

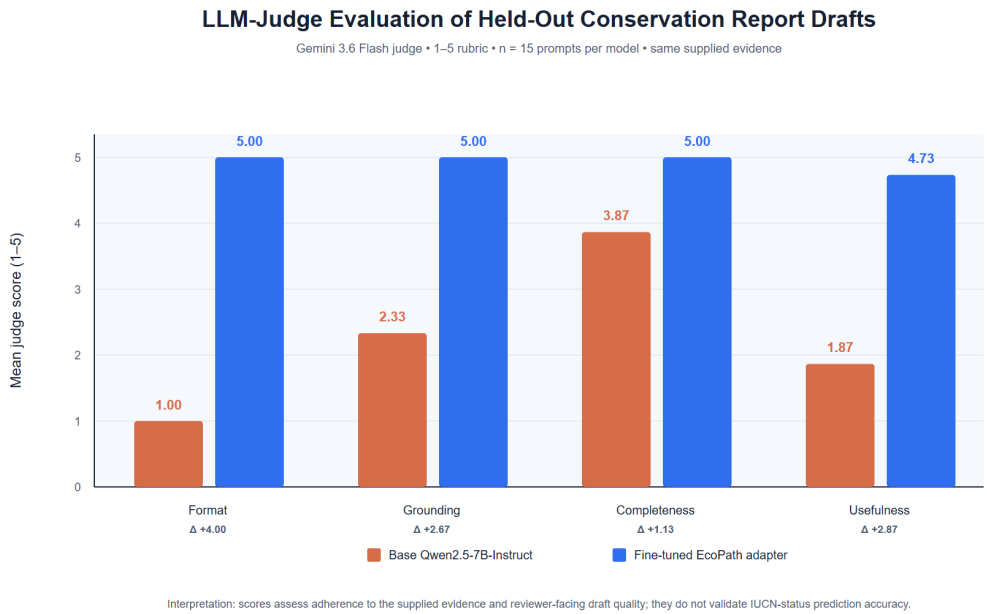


Figure 3. Mean automated rubric scores for the paired report conditions. Scores concern the supplied evidence and report requirements, rather than verified conservation-assessment accuracy. Error bars are unavailable.

4.3 Per-Species Results

The structural difference was consistent across the 15 held-out species: every unadapted response failed the checks, and every adapted response passed. Adapted reports were also shorter for each species. The species list and two illustrative report comparisons are provided in Appendix A.

5 Discussion

5.1 Interpreting the Structural and Rubric Gap

The comparison shows that a small synthetic training set was sufficient to establish the requested reporting format in this configuration. Within EcoPath, that change has a concrete application-level benefit: a structured response can be parsed and presented through a consistent interface. The result addresses the first research question, while leaving open whether the same reliability could be achieved with a stronger unadapted baseline.

The supplementary ratings suggest that the change extended beyond the visible format, because the judge could assess the unadapted Markdown responses as well as the adapted JSON. Higher grounding and usefulness scores are consistent with reports that state information gaps more directly and avoid generic elaboration. However, response structure, brevity, and similarity to the target style could also influence the ratings. The automated comparison therefore motivates closer examination of the content rather than establishing a measured reduction in unsupported claims.

The distinction becomes particularly important in this dataset. When the same ecological fields are missing throughout, a model can learn to mark them unavailable without learning when to report a supported value. What appears to be appropriate restraint under sparse inputs could become excessive abstention under richer ones. This is the central unresolved question raised by the findings: did the adapter learn a useful evidence-sensitive behaviour, a consistent template, or a combination of the two?

5.2 Qualitative Analysis

The Herring Gull (*Larus argentatus*) and Common Murre (*Uria aalge*) cases illustrate why the distinction matters. The unadapted reports introduced threats such as deforestation and wetland drainage without corresponding support in the input. The adapted reports instead identified missing information and retained the GBIF source reference. In the Common Murre case, the adapted response also explained that the available material could not support an IUCN-style assessment.

For a reviewer, the practical difference is between a plausible account that requires verification and an explicit account of what still needs to be established. The adapted examples better expressed the latter role. Nevertheless, two illustrative cases cannot show how often the same behaviour occurred across all factual statements. Source retention also establishes traceability only at the report level; it does not demonstrate that each claim is supported. Claim-level analysis is needed to connect these examples to an estimated error rate.

5.3 Relationship to Prior Work

Our findings fit the task-specific adaptation enabled by LoRA and QLoRA [4,6]. Their contribution here is not a new adaptation algorithm, but an application in which report structure and missing-information statements can be learned from a small dataset. Castro et al. [9] demonstrate the promise of prompt-based ecological information processing, making strong prompting a necessary next comparator for determining whether adaptation adds value in this setting.

The ecological benchmarks of Dorm et al. [12] and Uryu [13] place the result in a broader context. They evaluate knowledge and judgement tasks that differ from drafting a report from supplied metadata. EcoPath did not predict verified conservation categories or infer population trajectories, and its structural results should not be compared numerically with those ecological prediction scores. The relevant connection is the need to define the task narrowly enough that successful performance has a clear meaning.

Tam et al. [10] further motivate distinguishing format adherence from task success, while Zheng et al. [8] explain why automated judgements require independent scrutiny. Together, these studies support the next stage of evaluation: hold the output contract constant and test whether an adapted model makes better use of evidence than competitive alternatives. The current experiment establishes feasibility for that comparison without resolving it.

5.4 Role within the Wider EcoPath Platform

EcoPath connects information exploration with report preparation. Search, species profiles, and maps help users inspect available records; the drafting module offers a way to organise selected inputs for subsequent review. The structural improvement observed here supports that interface role. The study did not evaluate whether the reports reduced reviewer effort, improved decisions, or increased trust appropriately, and these outcomes remain separate application questions.

The most defensible use is therefore an initial draft that makes missing evidence visible. A report should help a reviewer identify the information available and the questions that remain, with suggested data-collection actions distinguished from species facts. Formal conservation interpretation continues to require appropriate evidence and expertise.

5.5 Limitations

The small, synthetic dataset limits the range of conclusions. Training and evaluation examples came from the same generation pipeline, and the split did not establish generalisation across taxonomic groups. Species-selection criteria were not recorded sufficiently to characterise sampling bias. Although the platform was motivated by Indian bird information needs, the held-out species did not constitute an India-specific evaluation. The findings therefore concern the sampled reporting examples rather than a regional conservation population.

The evidence design also limits the interpretation of missing-information handling. Verified status, trend, population, habitat, and range values were absent, so the experiment could not distinguish selective abstention from a repeated unavailable-field response. Controlled tests with both present and missing fields are needed to measure supported extraction, false abstention, and unsupported statements jointly.

The comparison included one unadapted prompting condition and one training run. Engineered prompts, schema constraints, and deterministic templates could produce different results. There was no independent validation set or repeated-seed analysis. Complete prompt, chat-template, decoding, model-revision, optimisation, and hardware details were not documented in the experimental materials available for this paper, limiting reproducibility and preventing a measured cost comparison.

Finally, automated scores were not validated against blinded human review, and teacher-judge independence was not established. The provider identifier corresponding to the recorded judge label, "Gemini 3.6 Flash", also requires verification. These limitations prevent interpreting the rubric as a calibrated measure of ecological correctness. Future evaluation should combine direct evidence checks with reviewer assessment and report errors in unsupported claims, ignored evidence, and source support.

6 Conclusion

We investigated QLoRA adaptation as a means of preparing structured bird conservation report drafts within EcoPath. On 15 held-out species, the adapted model consistently produced the requested JSON structure, whereas the unadapted condition returned Markdown prose. Adapted reports were shorter and received higher automated ratings, while illustrative cases showed explicit acknowledgement of evidence gaps.

The study establishes that the reporting pattern can be learned under the tested sparse-input conditions. It also identifies the next scientific question: whether adaptation improves evidence use once format is controlled and the availability of ecological information varies. Answering that question requires stronger baselines, direct claim and abstention measures, repeated runs, and human review.

For EcoPath, the immediate implication is a bounded drafting role. A useful report should organise what is known and make the remaining evidence needs easier to inspect. The present results support the feasibility of that workflow while leaving its broader factual reliability and practical value to be established.

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A Held-Out Evaluation Materials

Table 4 lists the 15 held-out species and their structural outcomes. Table 5 summarises the Herring Gull and Common Murre comparisons used to examine missing-information handling. The cases illustrate report behaviour and do not constitute a systematic estimate of unsupported-claim frequency.

Table 4. Structural outcomes for the 15 held-out species.

Held-out species	Base: JSON / schema	Adapted: JSON / schema
Herring Gull	Fail / Fail	Pass / Pass
Mediterranean Gull	Fail / Fail	Pass / Pass
European Goldfinch	Fail / Fail	Pass / Pass
Eurasian Hobby	Fail / Fail	Pass / Pass
Eurasian Hoopoe	Fail / Fail	Pass / Pass
Common Murre	Fail / Fail	Pass / Pass
Southern Cassowary	Fail / Fail	Pass / Pass
Crested Lark	Fail / Fail	Pass / Pass
Tawny Owl	Fail / Fail	Pass / Pass
Song Thrush	Fail / Fail	Pass / Pass
Common House-Martin	Fail / Fail	Pass / Pass
Peregrine Falcon	Fail / Fail	Pass / Pass
European Turtle-Dove	Fail / Fail	Pass / Pass
Eurasian Collared-Dove	Fail / Fail	Pass / Pass
Barn Swallow	Fail / Fail	Pass / Pass

Table 5. Illustrative paired reports under sparse evidence.

Species and supplied evidence	Unadapted output	Adapted output
Herring Gull (*Larus argentatus*). GBIF-linked identity and source reference; key ecological fields absent.	Markdown prose. Introduced deforestation without input support. Automated grounding score: 2/5.	Valid JSON. Marked missing fields unavailable, retained the source, and recommended data collection. Automated grounding score: 5/5.
Common Murre (*Uria aalge*). Same sparse evidence boundary.	Markdown prose. Introduced wetland drainage and deforestation without input support. Automated grounding score: 1/5.	Valid JSON. Stated that an IUCN-style assessment could not be completed and requested more evidence. Automated grounding score: 5/5.

The cases were not selected through a documented random sampling procedure. The reported pattern should therefore be interpreted alongside the aggregate structural results rather than as an estimated factual error rate.